CLASSIC VISUALS DASHBOARD OF USA SPENDING ON KANSAS HEALTH AND HUMAN SERVICES: 2020-2024

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Abstract:

This project titled **"USA Spending on Kansas Contracts & Grants (2020–2024)"** traces federal funding from the **U.S. Department of Health and Human Services (HHS)** to recipients across the state of Kansas and the Data is pulled from USAspending.gov, which is the U.S. government's official open-data portal.

The objective of the project is to transform giant amounts of federal contracts, grants data and visualize them as **interactive, business-friendly insights** using **Microsoft Power BI**. The dashboard is built in compliance with professional BI standards — with KPI cards, ranking views, trend charts, and region-level analysis — and visualization design guidelines and storytelling flow.

The project demonstrates proficiency in **data visualization, story-telling, and analytic reasoning** based on professional requirements for Power BI analysts.

Part 1: Data Selection & Analysis:

1.1 Dataset Selection:

Dataset which interests me are the ones which shows impact, efficiency, challenges or opportunities in our day-to-day lives and to make our community, society & Livelihood better with following data driven decision making. In this regard I prioritized working on USA Federal government spending in Kansas under Department of Health and Human Services (HHS) Agency. Hence I chose the following Project: **U.S. Health and Human Services (HHS) Spending in Kansas (2020 – 2024)**

For this Project, Data was downloaded from the official **USAspending.gov** portal, the U.S. government's open data platform for federal contracts and financial assistance (Grants) to the state of Kansas from 2020 – 2024, individually each year as per respective web page design.

Regular Separate CSV files were extracted for each year (2020–2024) covering both Grants and Contracts, then combined in Power BI using Power Query, since both Grants and Contracts, individually by themselves, have same columns/labels/fields.

1.2 Variable Identification:

After merging all **10 files** (Grants + Contracts for 2020 – 2024), the integrated dataset contains:

- **Rows:** 12,542 records
- **Columns:** ~7 core fields used for analysis out of many fields from both Grants & Contracts schemas.

Only these 7 fields are chosen which directly contributes to our planned DAX measures, visualizations, and relationships. Others are hidden in Power BI Model View.

Numerical variables:

Field Name	Description / Business Meaning
federal_action_obligation	Dollar amount obligated by the federal government for a given award(Grant) or Contract (transaction-level spending value).

Note: This Numerical Variable is further used to calculate and create various DAX measures.

Categorical variables:

Field Name	Description / Business Meaning
Category	Derived helper column identifying whether a record belongs to **Grant** or **Contract** (based on source file name).
recipient_name	Recipient organization or entity receiving the federal award.
recipient_county_name	County in Kansas where the recipient is located (used for spending by county visual).
primary_place_of_performance_city_name	City where funded work or service is performed.
Source_file	Original CSV source (e.g., HHS_GRANTS_2023.csv) used for audit and category classification.

Temporal variables:

Field Name	Description / Business Meaning
action_date	Date the transaction occurred (used for time-series and YOY analysis).
Calendar [Date] *(from Date Table)*	Continuous date dimensions are used to establish relationship with action_date.

Thus, the dataset integrates numerical (funding values), categorical (program and recipient details), and temporal (time-based trends) attributes needed for a complete Power BI finance dashboard story on USA spending patterns on HHS Contracts and Grants.

1.3 Expected Insights & DAX Measures:

DAX Measures:

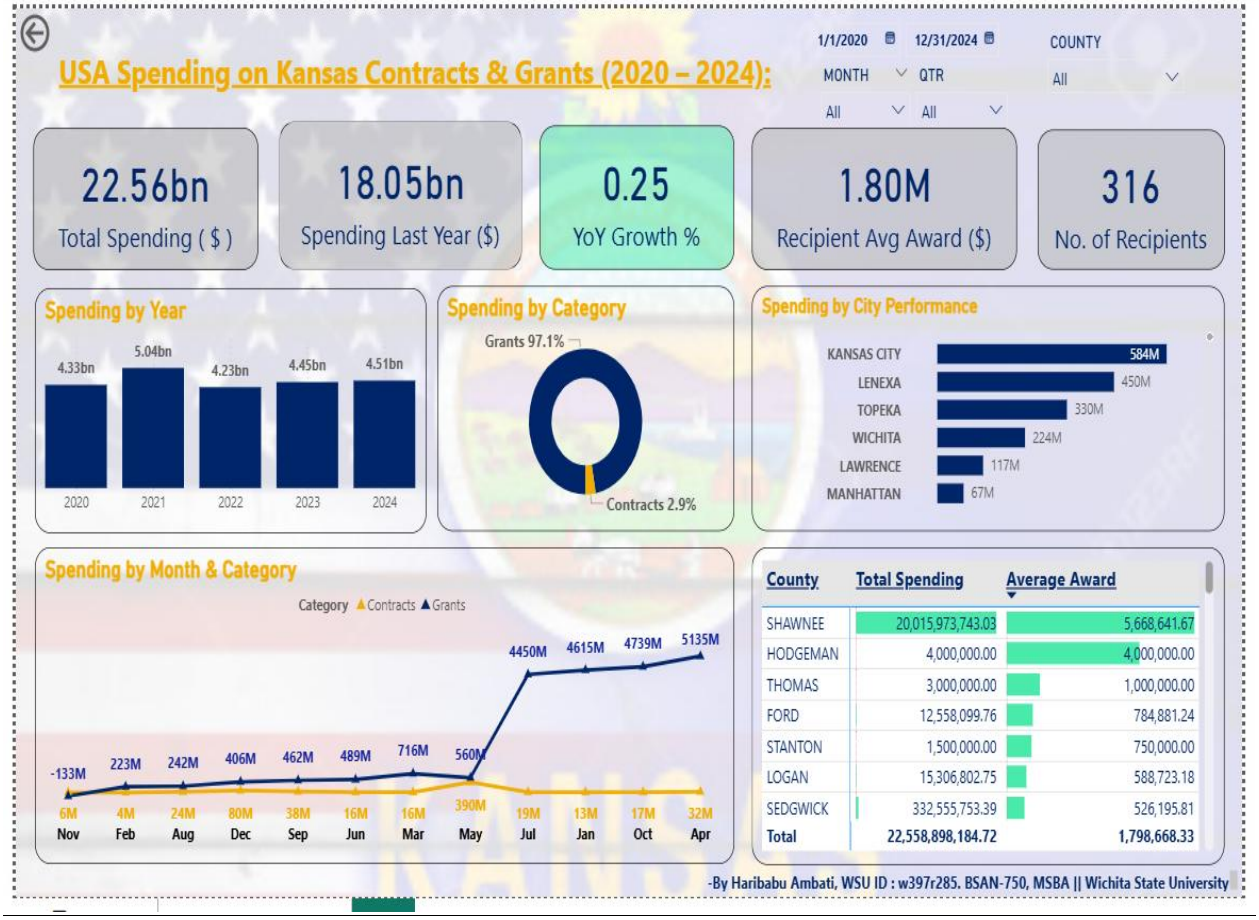
Measure Name	Formula:	Purpose:
Total Spending ($)	SUM(USAspending_HHS_Kansas[federal_action_obligation])	Calculates the total federal spending obligations in Kansas from 2020–2024.

Total Spending Last Year ($)	CALCULATE ([Total Spending ($)], DATEADD (Calendar [Date], -1, YEAR))	Compares current year's spending with previous year to analyze annual trend.
YoY Growth %	DIVIDE ([Total Spending ($)] - [Total Spending Last Year ($)), [Total Spending Last Year ($)])	Measures growth/decline in Kansas spending year over year growth.
Recipient Average Award ($)	Finds the average amount awarded per recipient, indicating funding scale.	Finds the average amount awarded per recipient, indicating funding scale.
Distinct Recipients	DISTINCTCOUNT(USAspending_HHS_Kansas[recipient_uei])	Counts how many unique organizations received funding.

Expected Insights:

- Identify **growth or contraction** in federal spending over time (Month & Year) in Kansas.
- Understand **distribution of funds** between **Contracts** and **Grants**.
- Evaluate **average award sizes** and **recipient numbers**.
- Recognize **which Kansas cities or counties** secured the highest HHS funding.
- Enable policymakers to view **funding effectiveness** across cities and time of years.

Part-2: Dashboard Construction:



Dashboard Overview:

The Power BI dashboard titled "USA Spending on Kansas Contracts & Grants (2020 – 2024)" is built in a 3 × 2 layout, combining KPIs, comparative visuals, and dynamic charts to present a clear story of how the U.S. Department of Health and Human Services (HHS) allocated federal funds to Kansas recipients.

This Dashboard design follows following flow:

Overview → Comparisons → Trends → Details.

2.1: Row 1 – KPI Cards (Performance Summary):

This row has KPI Card metrics –

- Total Spending ($)
- Total Spending Last Year ($)
- YoY Growth % &
- Recipient Avg Award ($)
- No. of Recipients

Note: The YoY Growth % KPI Card uses green (#49EAAA) and Red (#E68F96) conditional background formatting to highlight Positive & Negative growth performance, respectively.

2.2 Row 2 – Comparative Visuals (Composition & Ranking):

- ***Spending by Year-Column Chart (Ranking):***
  Highlights annual spending distribution from 2020–2024 and yearly growth trend except drastic growth in 2021 due is tragic post covid pandemic effect.

- ***Spending by Category-Donut Chart (Structure):***
  Shows composition of Grants vs Contracts (97.1% vs 2.9%).

- ***Spending by City Performance-Bar Chart (Ranking):***
  Kansas cities are ranked by total federal spending performance-wise.

2.3 Row 3 – Trend & Detailed Views:

- ***Spending by Month & Category-Line Chart (Time Series):***
  Shows monthly ascending order spending trend split by Grants and Contracts categories for 2020–24, where April month stands at High.

- ***Spending by County Matrix- Matrix with Conditional Formatting:***
  Displays each county's total spending and average award side by side. Bars within cells use green (Color code: #49EAAA) formatting to highlight higher values.

2.4 Filters / Slicers:

The dashboard consists of various interactive slicers that allow users to view the data and perform time/region-based exploration of federal spending patterns. These slicers update all visuals across the page to maintain synchronized filtering.

- ***Date Range Selector:***
  Users can select any date from the range between Jan 1, 2020 – Dec 31, 2024, to focus on a specific funding period.

- ***Month Selector:***
  Enables month-level comparison of spending trends, useful for seasonal analysis.

- ***Quarter Selector (Q1–Q4):***
  Groups spending by fiscal quarters to identify peak or low-spending periods.

- ***County Selector:***
  Filters all visuals to display data for a specific Kansas county, helping locate localized grant or contract distribution.

All slicers are set to cross-filter mode, ensuring that any selection dynamically updates all visuals and KPIs for a unified, interactive dashboard experience.

2.5 Canvas Theme & Formatting:

- **Background:** Combined U.S. and Kansas flags as canvas image, symbolizing federal–state collaboration.

Primary Colors:

- **Blue (#002569)** – Main visual and text color (from Kansas flag) is used for visual fonts.
- **Yellow (#F1AD02)** – Accent and label highlight color (from Kansas flag) is used for Title & other format areas.

Card Highlight: Green (#49EAAA) for performance metrics and conditional growth indicators.

Part- 3. Business Value & Insights:

3.1 Executive Summary:

This Power BI report, "USA Spending on Kansas Contracts & Grants (2020–2024)," depicts how U.S. Department of Health and Human Services (HHS) federal funds were disbursed in the state of Kansas for five years, from 2020 to 2024. It includes contracts and grants data into a clean picture of government expenditure for the benefit of healthcare, education, research, and community development projects.

This dashboard employs a 3×2 storyboarding-influenced design — KPI cards in the top row give a snapshot of high-level performance, mid-row charts show category breakdown and rank by city, and bottom-row includes time trends and county data. Altogether, the visualizations complete overall federal funding relevant at the local level within Kansas communities.

The dashboard provides a preview of how U.S. administration government expenditures by the HHS department flow into Kansas, enabling NGOs and non-profit organizations to easily forecast the trend of finances and plan budgets. Health centers, Medicare and Medicaid facilities, and behavioral health facilities can utilize the information to estimate resources available and advocate for additional funding where needed. From Government administration, state and federal policymakers can also consider when and under what conditions the budget is released so resources can be spent in an efficient manner and on the basis of community health and safety requirements.

Additionally, interactive slicers like Date, Month, Quarter, County, and Category allow analysts to select focused time spans or locations for closer look. This allows stakeholders such as policy analysts, financial planners, and government administrators to measure equity of funding, identify under-funded pockets, and optimize program planning.

Finally, this dashboard is both a platform of performance information and a transparency tool, responding to the greater American imperative to make data open. It takes raw spending data from USASpending.gov and forms it into an intelligible, actionable story of economic support, policy impact, and public benefit for Kansas.

3.2 Extracted Insights:

Insight 1:

Kansas received approximately $22.56 billion in total HHS spending between 2020 and 2024.

- ***Source***: KPI Card – Total Spending ($)
- ***Importance***: Reflects Kansas's strong federal funding footprint and how these allocations support state-level programs under the Department of Health and Human Services.
- ***Action***: Continue monitoring allocation efficiency to ensure sustained support for key community and health initiative and use average per year benchmark to claim or demand HHS budget for the State for Kansas.

Insight 2:

Grants made up 97.1% of total funds, while contracts accounted for only 2.9%.

- ***Source***: Donut Chart – Spending by Category
- ***Importance***: Shows Kansas's reliance on federally supported grant programs for public development.
- ***Action***: Encourage greater participation in federal contract opportunities to diversify funding streams and boost local operational projects.

Insight 3:

Spending peaked in 2021 at around $5.04 billion, then stabilized near $4.5 billion annually.

- ***Source***: Column Chart – Spending by Year
- ***Importance***: Indicates a post-pandemic recovery surge and subsequent stabilization in spending levels.
- ***Action***: Use this trend for budget forecasting and to plan emergency response strategies during future economic disruptions.

Insight 4:

Based on City Performances - Kansas City and Lenexa were the top-funded cities, receiving about $584 million and $450 million, respectively.

- ***Source***: Bar Chart – Spending by City Performance
- ***Importance***: Highlights urban concentration of federal funds and potential gaps across smaller or rural counties.
- ***Action***: Implement balanced allocation policies to ensure equitable access to funding in underserved regions.

Insight 5:

Shawnee County led with $20 billion in grants and an average award size of $5.67 million.

- ***Source***: Matrix – County Spending Details
- ***Importance***: Demonstrates effective management and utilization of federal funds at the county level.
- ***Action***: Use Shawnee County's model as a benchmark for improving administrative efficiency in other Kansas counties.

Conclusion and Future Scope:

Conclusion:

The Power BI dashboard effectively visualized Kansas HHS spending from 2020–2024, converting raw USASpending data into clear insights on funding trends, growth, and regional impact.

Future Scope:

Future work can enhance the model through optimized data joins and unions, integration of additional states and years, and automated data refresh for broader, more reliable federal-spending analysis.

Supporting Citations:

<u>Reference Textbook:</u>

Kolokolov, A., & Zelensky, M. (2021). *Data Visualization with Microsoft Power BI: How to Design Savvy Dashboards*. O'Reilly Media.

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